

Data Structures

Interview Cracker question bank

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Arrays and Strings

The most common interview substrate. Most "easy" and "medium" problems are an array or a string with a pattern applied to it.

[Easy] Q. What is the difference between an array and a dynamic array?

Answer. A **static array** is a fixed-size, contiguous block of memory. Its size is decided at creation and never changes. `int[10]` in C or Java.

A **dynamic array** wraps a static array and grows it when it fills up, by allocating a larger block and copying. Python `list`, Java `ArrayList`, C++ `vector`, Go `slice`, JavaScript `Array`.

Both give $O(1)$ indexed access, because the address of element i is just $\text{base} + i \times \text{element_size} - \text{one}$ multiplication and one addition, no traversal.

Operation	Cost	Why
Access by index	$O(1)$	Direct address arithmetic
Append	$O(1)$ amortised	Occasional doubling + copy
Insert/delete at front or middle	$O(n)$	Everything after must shift
Search (unsorted)	$O(n)$	No structure to exploit
Search (sorted)	$O(\log n)$	Binary search

[Easy] Q. Why is inserting at the front of an array $O(n)$ but appending $O(1)$?

Answer. Because arrays are contiguous. Inserting at index 0 means every existing element must move one slot to the right to make room - n moves. Appending writes into already-reserved space at the end - one move.

This is the single most useful performance fact about arrays, and it is why:

- `list.pop(0)` is a performance bug in a loop; use `collections.deque`.
- Building a result by `insert(0, x)` is $O(n^2)$; append and reverse at the end.

[Medium] Q. Why are strings immutable in Python, Java and JavaScript?

Answer. Several reasons reinforce each other:

- **Safe sharing.** An immutable string can be passed around, cached and shared
- **Hashability.** A string's hash can be computed once and cached. If strings
- **Interning.** Identical literals can share one object, saving memory.
- **Security.** A filename or URL validated and then passed to the OS cannot be

The cost is that every "modification" allocates. Concatenating in a loop is $O(n^2)$; use a builder (`StringBuilder`, `"".join`, `array + join`) instead.

[Medium] Q. What does "in-place" mean, and what is the space complexity of an in-place algorithm?

Answer. In-place means the algorithm transforms the input using only $O(1)$ extra space - a few variables, no second array proportional to n .

```
def reverse_in_place(arr):          # O(1) extra space
    left, right = 0, len(arr) - 1
    while left < right:
        arr[left], arr[right] = arr[right], arr[left]
        left += 1
        right -= 1
```

Two honest caveats interviewers probe for:

- You cannot do this to a string in Python/Java - strings are immutable, so you
- $O(\log n)$ stack space from recursion is usually still called "in-place" by

[Medium] Q. What is the two-pointer technique and when does it apply?

Answer. Two indices moving through the array under a rule, replacing a nested loop - turning $O(n^2)$ into $O(n)$.

Three shapes:

Converging (from both ends) - needs a **sorted** array or a symmetry:

```
left, right = 0, len(arr) - 1
while left < right:
    total = arr[left] + arr[right]
    if total == target: return (left, right)
    if total < target: left += 1      # need bigger
    else:              right -= 1     # need smaller
```

This works only because sorting guarantees which direction increases the sum.

Fast/slow - cycle detection, finding the middle of a linked list.

Same direction (read/write) - in-place filtering, deduplication:

```
write = 0
for read in range(len(arr)):
    if keep(arr[read]):
        arr[write] = arr[read]
        write += 1
return write          # new logical length
```

The tell: the array is sorted, or you are looking for a pair/triplet, or you must filter in place.

[Medium] Q. What is the sliding window technique?

Answer. A window `[left, right]` over a contiguous run, where `right` expands and `left` contracts to restore a constraint. Each element enters and leaves at most once, so the whole scan is $O(n)$ despite the nested-looking loops.

```
def longest_unique(s):
    last_seen = {}
    left = best = 0
    for right, ch in enumerate(s):
        if ch in last_seen and last_seen[ch] >= left:
            left = last_seen[ch] + 1      # contract past the duplicate
        last_seen[ch] = right
        best = max(best, right - left + 1)
    return best
```

The tell: "longest/shortest/maximum **contiguous** subarray or substring satisfying X". If the problem says subsequence (non-contiguous), a window does not apply - that is usually dynamic programming.

[Medium] Q. How do you find a duplicate in an array? Give three approaches.

Answer. The interviewer wants to see you trade time against space.

- 1 **Brute force** - compare every pair. $O(n^2)$ time, $O(1)$ space.
- 2 **Sort first** - duplicates become adjacent. $O(n \log n)$ time, $O(1)$ extra space
- 3 **Hash set** - one pass, check membership. $O(n)$ time, $O(n)$ space.

Then the constrained follow-up: values are in $1 \dots n$ and you may not modify the input or use extra space. That is Floyd's cycle detection applied to the array as an implicit linked list ($i \rightarrow \text{arr}[i]$), giving $O(n)$ time and $O(1)$ space.

Say all three and name the trade-off explicitly - that is the answer they want, not just the fastest one.

[Easy] Q. What is the difference between a subarray, a subsequence and a subset?

Answer. A distinction that changes which algorithm applies, so get it right:

Term	Contiguous?	Order kept?	Count for n elements
Subarray / substring	Yes	Yes	$n(n+1)/2$
Subsequence	No	Yes	2^n
Subset	No	No	2^n

For $[1, 2, 3]$: $[2, 3]$ is a subarray; $[1, 3]$ is a subsequence but not a subarray; $\{3, 1\}$ is a subset.

Why it matters: subarray problems usually yield to a sliding window or prefix sums ($O(n)$). Subsequence problems usually need dynamic programming, because there are exponentially many.

[Medium] Q. What is a prefix sum and what is it for?

Answer. A precomputed array where `prefix[i]` is the sum of the first i elements. It makes any range sum $O(1)$:

```
prefix = [0]
for n in nums:
    prefix.append(prefix[-1] + n)

# sum of nums[i:j] in O(1)
range_sum = prefix[j] - prefix[i]
```

Build cost $O(n)$ once, then unlimited $O(1)$ range queries - the right trade when you answer many queries over a static array. Combined with a hash table of seen prefix sums it solves "count subarrays summing to k" in one pass.

Worked examples: [coding-challenges/two-pointers/](#) and [coding-challenges/sliding-window/](#).

Hash Tables

The most useful data structure in interviews. A very large fraction of "optimise this $O(n^2)$ solution" problems are solved by adding a hash table.

[Easy] Q. How does a hash table work?

Answer. It stores key-value pairs in an array, and uses a hash function to compute which slot a key belongs in.

- 1 `hash(key)` produces an integer.
- 2 That integer is reduced to an index, usually `hash % capacity`.
- 3 The pair is stored at that index.

Lookup repeats the same computation, so it goes straight to the slot instead of scanning - $O(1)$ average instead of $O(n)$.

The whole structure rests on one assumption: that the hash function spreads keys roughly evenly. Everything that goes wrong with hash tables is that assumption failing.

[Easy] Q. What is a collision, and how are collisions resolved?

Answer. A collision is two different keys mapping to the same slot. They are unavoidable - there are more possible keys than slots (the pigeonhole principle).

Two families of solution:

Separate chaining. Each slot holds a list of entries. On collision, append. Lookup scans that one short list. Simple, degrades gracefully, but costs a pointer per entry and scatters memory. Used by Java's `HashMap` (which upgrades a long chain to a balanced tree) and most C++ `unordered_map` implementations.

Open addressing. Everything lives in the array itself. On collision, probe for another free slot - linearly, quadratically, or via a second hash. Better cache behaviour, no per-entry allocation, but deletion is awkward (you must leave a tombstone, or later probes stop too early) and it degrades badly when full. Used by Python's `dict` and Go's maps.

[Medium] Q. Hash table lookup is " $O(1)$ ". When is it not?

Answer. $O(1)$ is the average case. The worst case is $O(n)$, when every key hashes to the same slot and the table degenerates into a linked list.

This matters in practice in two ways:

- **Bad hash functions.** If you write `hashCode()` as `return 1`, every lookup
- **Hash-flooding attacks.** If an attacker can choose your keys and knows your

Java 8+ mitigates it differently: a bucket whose chain exceeds 8 entries converts to a red-black tree, capping the worst case at $O(\log n)$.

[Medium] Q. What is the load factor and why does it trigger a resize?

Answer. Load factor = entries / capacity. It measures how full the table is.

As it rises, collisions rise, and average probe length rises with it. Past roughly 0.7 the performance falls off sharply, so implementations resize - double the capacity and **rehash every entry**, because the index depends on capacity.

That rehash is $O(n)$, which makes a single insert occasionally $O(n)$ - but amortised $O(1)$ across all inserts, for exactly the same geometric-growth reason a dynamic array append is.

Practical consequence: if you know you will insert a million entries, pre-size the map. You avoid about 20 rehashes.

[Medium] Q. What makes a good hash function?

Answer. Three properties:

- 1 **Deterministic** - the same key always hashes to the same value. Within one
- 2 **Uniform** - outputs spread evenly across the range, so no slot is favoured.
- 3 **Fast** - it runs on every single operation.

Note that cryptographic strength is not required and usually not wanted; SHA-256 is uniform but far too slow for a hash table.

[Hard] Q. What is the contract between equals and hashCode?

Answer. If two objects are equal, they must have the same hash code. The converse is not required - unequal objects may share a hash code, that is just a collision.

Break it and hash tables break silently:

```
Set<Point> set = new HashSet<>();
set.add(new Point(1, 2));
set.contains(new Point(1, 2)); // false, if hashCode() was not overridden
```

The lookup hashes to a different slot and never finds the entry, even though `equals` would have said yes.

The other half of the contract: **a key's hash must not change while it is in the table**. Mutating a field that feeds `hashCode` strands the entry in the wrong slot - it is in the table but unreachable, and it will not even be removed by `remove()`. This is why immutable keys are strongly preferred, and why Python simply refuses to hash a `list`.

[Medium] Q. When would you not use a hash table?

Answer. Four cases:

- **You need ordering.** Hash tables have none. Use a balanced BST (`TreeMap`,
 - **You need the worst case bounded.** Real-time systems cannot accept an
 - **The dataset is tiny.** For five entries, a linear scan of an array is faster
 - **Keys are not hashable.** Mutable or unorderable keys.
-

[Medium] Q. What is the difference between a HashMap, a HashSet and a HashTable?

Answer.

- **Map** - key -> value. The core structure.
 - **Set** - keys only. Almost always implemented as a map with a dummy value.
 - **Hashtable** (the legacy Java class, capital-T) - a synchronised, slower
-

[Medium] Q. How do you use a hash table to turn $O(n^2)$ into $O(n)$?

Answer. The pattern: **replace an inner search loop with a lookup**, trading memory for time.

Two Sum is the canonical example. The brute force checks every pair:

```
for i in range(len(nums)):          # O(n^2)
    for j in range(i + 1, len(nums)):
        if nums[i] + nums[j] == target:
            return [i, j]
```

The hash table version asks, for each number, "have I already seen the value that completes this pair?" - a lookup instead of a scan:

```
seen = {}                          # value -> index
for i, n in enumerate(nums):        # O(n) time, O(n) space
    if target - n in seen:
        return [seen[target - n], i]
    seen[n] = i
```

Recognising this - "I am searching inside a loop, so I should index instead" - is worth more than memorising any individual solution. Worked examples are in [coding-challenges/hashing/](#).

Linked Lists

Rarely the right choice in production, reliably popular in interviews - because they test pointer manipulation and careful edge-case handling with no library to hide behind.

[Easy] Q. What is a linked list and how does it differ from an array?

Answer. A chain of nodes, each holding a value and a reference to the next node. Nodes live anywhere in memory; the links impose the order.

	Array	Linked list
Access by index	O(1)	O(n) - must walk
Insert/delete at head	O(n)	O(1)
Insert/delete at a known node	O(n)	O(1)
Memory per element	value only	value + pointer(s)
Cache locality	Excellent	Poor
Size	grows by reallocation	grows one node at a time

The honest summary: linked lists win only when you are inserting or deleting at a position you already hold a reference to, and never index. In practice, cache locality means an array often wins even then - a linear scan of a contiguous array can outrun pointer-chasing by an order of magnitude.

[Easy] Q. Singly vs doubly linked - what does the extra pointer buy?

Answer. A `prev` pointer costs one word per node and buys:

- **Backwards traversal.**
- **O(1) deletion given only the node** - a singly linked list needs the
- **O(1) append with a tail pointer** - also possible singly, but removing the

This is why an LRU cache uses a doubly linked list: it must move an arbitrary node to the front in O(1), which requires unlinking it, which requires `prev`.

[Medium] Q. How do you detect a cycle in a linked list?

Answer. Floyd's tortoise and hare. Two pointers, one moving one step per iteration and one moving two. If there is a cycle they must eventually meet; if there is not, the fast pointer reaches the end.

```
def has_cycle(head):
    slow = fast = head
    while fast and fast.next:
        slow = slow.next
        fast = fast.next.next
    if slow is fast:
        return True
    return False
```

O(n) time, O(1) space. The hash-set alternative (store every visited node) is also O(n) time but O(n) space - mention both, then say why you would pick Floyd.

Why they must meet: once both are inside the cycle, the fast pointer gains exactly one position per iteration, so it closes any gap of size k in k steps. It cannot "jump over" the slow pointer.

The follow-up - find where the cycle starts. After they meet, reset one pointer to the head and advance both one step at a time. They meet at the cycle entrance. (The distance from head to entrance equals the distance from the meeting point to the entrance, going forward around the loop.)

[Medium] Q. Reverse a singly linked list.

Answer. Three pointers, walking once and flipping each link as you pass.

```
def reverse(head):
    prev = None
    curr = head
    while curr:
        next_node = curr.next    # save it BEFORE overwriting
        curr.next = prev        # flip
        prev = curr              # advance
        curr = next_node
    return prev                  # new head
```

O(n) time, O(1) space. The whole problem is saving `curr.next` before you destroy it - do that in the wrong order and you lose the rest of the list.

The recursive version is elegant but **O(n) space** from the call stack, which is a real downside worth naming rather than presenting it as equally good.

[Medium] Q. Find the middle of a linked list in one pass.

Answer. Same fast/slow trick. When the fast pointer reaches the end, the slow pointer is at the middle.

```
def middle(head):
    slow = fast = head
    while fast and fast.next:
        slow = slow.next
        fast = fast.next.next
    return slow
```

For even lengths this returns the second of the two middles. If you need the first, stop when `fast.next` and `fast.next.next`. Interviewers often care which one you return - say out loud which convention you chose.

[Medium] Q. Why does so much linked-list code use a dummy head node?

Answer. Because it removes the "what if the head changes?" special case.

Without a dummy, deleting the first node or inserting before it needs separate branching, and that branch is where the bug always is. With a sentinel node in front, every node has a predecessor and one uniform code path handles all of them:

```
def remove_all(head, target):
    dummy = Node(None)
    dummy.next = head
    prev = dummy
    while prev.next:
        if prev.next.value == target:
            prev.next = prev.next.next    # works even for the real head
        else:
            prev = prev.next
    return dummy.next                    # not `head` - it may be gone
```

Reaching for a dummy node unprompted signals real experience.

[Hard] Q. Merge two sorted linked lists. What is the complexity, and why is it nicer than for arrays?

Answer. Walk both with a pointer each, always taking the smaller head, and splice it onto a result list built behind a dummy node.

$O(n + m)$ time, $O(1)$ extra space - and that $O(1)$ is the interesting part. The array version of merge needs an $O(n + m)$ output buffer, because you cannot make room in a contiguous block without copying. A linked list merge just re-points existing nodes, allocating nothing.

This is the one place linked lists genuinely beat arrays, and it is why merge sort on a linked list is $O(1)$ -space while merge sort on an array is $O(n)$ -space.

A worked implementation is in `examples/java/algorithms/merge_two_sorted_lists.java` and `examples/c/data-structures/linked_list.c`.

[Medium] Q. What edge cases must linked-list code handle?

Answer. Interviewers grade heavily on these, and they are the same four every time:

- 1 **Empty list** (`head` is `None`).
- 2 **Single node** (`head.next` is `None`) - many loops never execute.
- 3 **Operating on the head** - does your code return the right new head?
- 4 **Operating on the tail** - is `next` set to `None`, or did you leave a

Walk through a 0-node, 1-node and 2-node list out loud before you say you are done. It takes twenty seconds and catches most bugs.

Stacks, Queues and Heaps

Three small structures that unlock a surprising number of problems once you learn to recognise their signals.

[Easy] Q. Stack vs queue?

Answer. Both restrict where you may add and remove; they differ in which end.

- **Stack** - LIFO, last in first out. `push/pop` at the same end.
- **Queue** - FIFO, first in first out. Add at the back, remove from the front.

All four operations are $O(1)$ in a good implementation.

Where each shows up:

Stack	Queue
Function call stack	BFS / level-order traversal
Undo history	Task and message queues
Bracket matching	Print spoolers, request buffers
DFS (explicit or via recursion)	Rate limiting, buffering
Expression evaluation	Producer-consumer pipelines

The rule of thumb: DFS uses a stack, BFS uses a queue. That one fact decides a lot of problems.

[Easy] Q. How do you implement a queue efficiently?

Answer. Not with a plain array and `pop(0)` - that shifts every element, so it is $O(n)$ per dequeue and $O(n^2)$ overall.

Three correct options:

- **Doubly linked list** - $O(1)$ at both ends, but a pointer per node and poor
- **Circular buffer** - a fixed array with `head` and `tail` indices that wrap via
- **Deque from the standard library** - `collections.deque`, `ArrayDeque`,

The classic puzzle - implement a queue with two stacks - is worth knowing: push onto `in`; to pop, if `out` is empty pour all of `in` into it (reversing order) then pop from `out`. Each element moves at most twice, so it is **$O(1)$ amortised**.

[Medium] Q. What is a monotonic stack and what does it solve?

Answer. A stack kept in sorted order - you pop elements that break the order before pushing. It answers "for each element, what is the next element greater (or smaller) than it?" in **$O(n)$** instead of $O(n^2)$.

```
def next_greater(nums):
    result = [-1] * len(nums)
    stack = []  # holds indices, values decreasing
    for i, n in enumerate(nums):
        while stack and nums[stack[-1]] < n:
            result[stack.pop()] = n  # n is the answer for that index
        stack.append(i)
    return result
# [2, 1, 3] -> [3, 3, -1]
```

Each index is pushed once and popped once, so despite the inner `while` the total work is $O(n)$.

The tell: "next greater / previous smaller element", "largest rectangle in a histogram", "daily temperatures", "trapping rain water". If a problem asks you to relate each element to the nearest element satisfying a comparison, try a monotonic stack.

[Easy] Q. What is a heap, and what is it good at?

Answer. A complete binary tree satisfying the heap property: in a min-heap, every parent is $\leq$ its children (so the minimum sits at the root).

It is stored in a **flat array**, not with pointers - for index i , children are at $2i+1$ and $2i+2$, parent at $(i-1)//2$. That is why heaps are fast: no allocation, excellent cache locality.

Operation	Cost
Peek min/max	$O(1)$
Insert	$O(\log n)$ - sift up
Extract min/max	$O(\log n)$ - sift down
Build from n items	$O(n)$, not $O(n \log n)$
Search for arbitrary value	$O(n)$ - no ordering between siblings

Two points interviewers probe:

- **Building a heap is $O(n)$.** Sifting down from the middle backwards costs less
 - **A heap is only partially ordered.** Finding an arbitrary element is $O(n)$, and
-

[Medium] Q. How do you find the k largest elements in a stream of n items?

Answer. A min-heap of size k .

```
import heapq

def k_largest(stream, k):
    heap = []                                # min-heap of the best k so far
    for item in stream:
        if len(heap) < k:
            heapq.heappush(heap, item)
        elif item > heap[0]:                 # bigger than the smallest kept
            heapq.heapreplace(heap, item)
    return sorted(heap, reverse=True)
```

$O(n \log k)$ time, $O(k)$ space.

The counter-intuitive part, and the point of the question: to track the k largest you use a **min-heap**. The root is the weakest of your current champions, so it is both the cheapest to check against and the right one to evict.

Why not sort? Sorting is $O(n \log n)$ time and **$O(n)$ space** - impossible if the stream does not fit in memory. When k is small and n is huge, $O(n \log k)$ with $O(k)$ space is the meaningful win.

[Medium] Q. What is a priority queue and how does it differ from a heap?

Answer. A **priority queue** is an abstract data type: "give me the highest-priority item next". A **heap** is the usual concrete implementation.

The distinction matters because other implementations exist with different trade-offs - a sorted array gives $O(1)$ extraction but $O(n)$ insertion; a Fibonacci heap gives $O(1)$ amortised insert and decrease-key, which is what makes the textbook Dijkstra bound $O(E + V \log V)$.

In interviews, "priority queue" almost always means "binary heap". It is the engine inside Dijkstra's algorithm, A*, Huffman coding, and event-driven simulation.

[Medium] Q. How do you check for balanced brackets?

Answer. The canonical stack problem. Push opening brackets; on a closing bracket, the top of the stack must be its partner.

```
PAIRS = {"(": ")", "[": "]", "{": "}"

def is_balanced(s):
    stack = []
    for ch in s:
        if ch in "([{":
            stack.append(ch)
        elif ch in PAIRS:
            if not stack or stack.pop() != PAIRS[ch]:
                return False          # wrong partner, or nothing open
    return not stack                  # anything left open is unbalanced
```

$O(n)$ time, $O(n)$ space. The two edge cases that separate a correct answer from a nearly-correct one: a closing bracket when the stack is **empty**, and leftover openers at the **end**. Candidates routinely forget the second and return `True` for `"((("`.

Trees and Graphs

The highest-yield topic after hash tables. Most "hard" interview problems are a graph problem wearing a costume.

[Easy] Q. What is a binary search tree, and what is its complexity?

Answer. A binary tree where, for every node, all keys in the left subtree are smaller and all keys in the right subtree are larger. That invariant makes search a series of "left or right?" decisions.

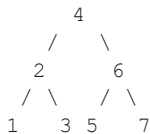
Search, insert and delete are **$O(h)$** where h is the height - **not** $O(\log n)$. The two coincide only when the tree is balanced:

- Balanced: $h \sim \log n \rightarrow O(\log n)$.
- Degenerate (insert already-sorted data): $h = n \rightarrow O(n)$, a linked list with

That degeneration is the entire reason self-balancing trees exist. Saying " $O(h)$, which is $O(\log n)$ only if balanced" is a much stronger answer than " $O(\log n)$ ".

[Easy] Q. What are the tree traversals and when do you use each?

Answer. Three depth-first orders, defined by when you visit the node relative to its children, plus one breadth-first order.

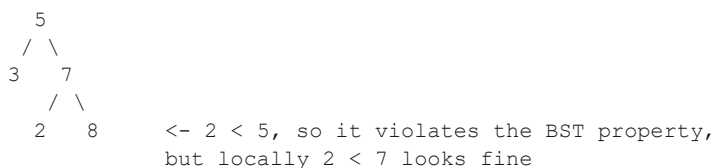


Traversal	Order	Result	Use
In-order	left, node , right	1 2 3 4 5 6 7	Sorted output from a BST
Pre-order	node , left, right	4 2 1 3 6 5 7	Copying/serialising a tree
Post-order	left, right, node	1 3 2 5 7 6 4	Deleting; computing from children up
Level-order (BFS)	by depth	4 2 6 1 3 5 7	Shortest path; level-by-level work

The fact worth memorising: **in-order traversal of a BST yields sorted order**. "Validate a BST" and "find the k th smallest" are both just that observation.

[Medium] Q. How do you validate a binary search tree?

Answer. The trap: checking only `left < node < right` locally is **wrong**. It passes trees that are not BSTs:



Every node must fall inside a range inherited from its ancestors:

```
def is_bst(node, low=float("-inf"), high=float("inf")):
```

```

if node is None:
    return True
if not (low < node.value < high):
    return False
return (is_bst(node.left, low, node.value) and
        is_bst(node.right, node.value, high))

```

Equivalently: do an in-order traversal and check the output is strictly increasing. Both are $O(n)$ time, $O(h)$ space.

[Medium] Q. Which balanced trees should I know, and what balances them?

Answer. You need the idea, not the rotation code.

- **AVL** - strictly balanced; heights of sibling subtrees differ by at most 1.
- **Red-black** - loosely balanced via colour rules; the longest path is at most
- **B-tree / B+ tree** - high branching factor, so the tree is very shallow. Used

All give $O(\log n)$ worst case. The distinction is the constant factor and the read/write balance.

[Easy] Q. BFS or DFS - how do you choose?

Answer.

	BFS	DFS
Structure	Queue	Stack (or recursion)
Visits	Nearest first	Deepest first
Finds shortest path	Yes , in an unweighted graph	No
Space	$O(\text{width})$ - can be huge	$O(\text{depth})$
Natural for	Levels, shortest hops, "minimum steps"	Cycles, topological sort, connectivity, backtracking

Both are $O(V + E)$ time.

The decisive rule: if the question says shortest, fewest, or minimum number of steps in an unweighted graph, use **BFS**. BFS reaches every node by the fewest edges, so the first time it arrives is the shortest way. DFS gives no such guarantee - it may find a long path first.

If edges are weighted, neither works: use **Dijkstra** (non-negative weights) or **Bellman-Ford** (negative weights allowed).

[Medium] Q. How do you represent a graph, and which representation should you pick?

Answer. Two options, chosen by density.

Adjacency list - map each vertex to its neighbours.

```
graph = {"a": ["b", "c"], "b": ["d"], "c": [], "d": []}
```

Space $O(V + E)$. Iterating a vertex's neighbours is $O(\text{degree})$. **The default.**

Adjacency matrix - a $V \times V$ grid where $m[i][j]$ marks an edge. Space $O(V^2)$ regardless of edge count. Edge lookup is $O(1)$; listing neighbours is $O(V)$.

Pick the matrix when the graph is **dense** (E approaches V^2), when you need constant-time "is there an edge?", or for matrix algorithms like Floyd-Warshall. Pick the list otherwise - real graphs are almost always sparse, and $O(V^2)$ space for a million vertices is a terabyte.

[Medium] Q. Why must graph traversal track visited nodes when tree traversal need not?

Answer. Because a tree is by definition acyclic and each node has exactly one parent - you cannot arrive anywhere twice. A general graph can have cycles and multiple paths to the same node, so without a `visited` set traversal loops forever or does exponential redundant work.

```
def dfs(graph, node, visited=None):
    if visited is None:
        visited = set()
    if node in visited:
        return
    visited.add(node)
    for neighbour in graph[node]:
        dfs(graph, neighbour, visited)
```

Mark a node **when you enqueue/visit it**, not when you dequeue it - otherwise the same node can be queued many times before it is ever processed.

[Hard] Q. What is a topological sort and when is it impossible?

Answer. A linear ordering of a directed graph's vertices such that every edge points forward - every task appears after everything it depends on. Build systems, task schedulers and course prerequisites all need it.

It is impossible exactly when the graph contains a cycle, since a cycle means a set of tasks that each depend on the other. So a topological sort doubles as a cycle detector, which is often the real question.

Kahn's algorithm, $O(V + E)$:

- 1 Compute every vertex's in-degree.
- 2 Enqueue all vertices with in-degree 0.
- 3 Pop one, append it to the output, and decrement its neighbours' in-degrees;
- 4 If the output holds fewer than V vertices, **there is a cycle**.

A recursive example over a hierarchy is in `examples/sql/recursive-ctes/org_chart_tree_traversal.sql`.

[Medium] Q. What is a trie and why use one over a hash table?

Answer. A tree keyed by character, where a word's letters spell out the path from the root. Shared prefixes share nodes.

Lookup is **$O(k)$** for a key of length k - independent of how many words are stored, and with no hashing and no collisions.

The reason to choose it over a hash map is **prefix queries**. "All words starting with `pre`" is one walk to the `pre` node followed by a subtree traversal. A hash map cannot do this at all without scanning every key, because hashing deliberately destroys the relationship between similar keys.

That makes tries the structure behind autocomplete, spell-checkers and IP routing tables. The cost is memory: a node per character per distinct prefix.

Tries & Prefix Data Structures

A trie (prefix tree) stores strings so that shared prefixes are stored once. It is the go-to structure when the question mentions "prefix", "autocomplete", or "dictionary word" - and it is the backbone of command completion, spell-checking, and routing tables.

[Easy] Q. What is a trie, and what does it buy you over a hash set of words?

Answer. A trie is a tree where each edge is labelled with a character and a path from the root spells a string. What it buys you is **prefix awareness**: `search`, `insert`, and `starts_with` all cost $O(L)$ in the string length L (not the number of words), and you can enumerate every word with a given prefix by walking the subtree - something a hash set cannot do at all.

```
class TrieNode:
    def __init__(self):
        self.children = {}
        self.is_end = False

class Trie:
    def __init__(self):
        self.root = TrieNode()

    def insert(self, word: str) -> None:
        node = self.root
        for ch in word:
            node = node.children.setdefault(ch, TrieNode())
        node.is_end = True

    def _walk(self, s: str):
        node = self.root
        for ch in s:
            if ch not in node.children:
                return None
            node = node.children[ch]
        return node

    def search(self, word: str) -> bool:
        node = self._walk(word)
        return node is not None and node.is_end

    def starts_with(self, prefix: str) -> bool:
        return self._walk(prefix) is not None

t = Trie()
for w in ["cat", "car", "card", "care"]:
    t.insert(w)
assert t.search("cat") and t.search("card") and not t.search("ca")
assert t.starts_with("car") and not t.starts_with("dog")
```

The key detail: `search("ca")` is `False` even though `starts_with("ca")` is `True`, because `ca` is a prefix but was never itself inserted - the `is_end` flag distinguishes "a word ends here" from "a path passes through here".

[Medium] Q. How do you list all words that share a prefix, and what's the cost?

Answer. Walk to the node for the prefix ($O(L)$), then a DFS of that subtree collecting every `is_end` node. The cost is $O(L + K)$ where K is the total number of characters in all matching words - you pay for the output, which is unavoidable.

```
def words_with_prefix(self, prefix: str) -> list[str]:
    node = self._walk(prefix)
    if node is None:
```

```

        return []
    out = []
    def dfs(n, acc):
        if n.is_end:
            out.append(prefix + acc)
        for ch in sorted(n.children):
            dfs(n.children[ch], acc + ch)
    dfs(node, "")
    return out

t.words_with_prefix("ca")    # ['car', 'card', 'care', 'cat']
t.words_with_prefix("do")    # []

```

This is exactly the autocomplete step. Note the `sorted(...)` makes the output deterministic (alphabetical); drop it if you only care about existence and want to stop early.

[Medium] Q. Trie vs sorted array + binary search vs hash set - when is each right?

Answer.

Need	Best fit	Why
"Is this exact word present?"	Hash set	O(L) expected, tiny constant
"Any word with this prefix?"	Trie	Subtree walk; the other two can't
"Nearest word / suggest close match"	Trie	DFS with a small edit budget
Memory is tight, no prefix queries	Sorted array	No per-node object overhead

A hash set wins on plain membership and uses far less memory. A trie's cost is one node object per distinct prefix, so a dictionary of 100k long words can be memory-heavy - which is why production systems often use a **radix trie** (compress unary chains into a single labelled edge) or a sorted array with binary search when they don't actually need prefix enumeration.

[Hard] Q. How would you implement "longest word in the dict that is a

prefix of `s`", and what's the complexity?

Answer. Insert every dictionary word into a trie, then walk `s` character by character, tracking the deepest `is_end` node you pass. When the walk falls off a branch, stop - you've found the longest prefix that exists. Time is $O(L + \text{total dict characters})$ to build plus $O(L)$ to query; space $O(\text{total dict characters})$.

```

def longest_prefix(trie: "Trie", s: str) -> str:
    node = trie.root
    best = ""
    for i, ch in enumerate(s):
        if ch not in node.children:
            break
        node = node.children[ch]
        if node.is_end:
            best = s[: i + 1]
    return best

t = Trie()
for w in ["we", "word", "flower", "flowering"]:
    t.insert(w)
assert longest_prefix(t, "flowerpowers") == "flower"
assert longest_prefix(t, "dog") == ""

```

The trap is returning the last matched character instead of the last position where a **complete word** ended - "flowe" walks fine but isn't a word, so `is_end` is what makes the answer "flower", not "flowe".

[Easy] Q. Why do we mark a terminal node instead of, say, storing the word?

Answer. Storing the full word at each terminal node would duplicate every prefix's characters and blow the space from $O(\text{total characters})$ to $O(\text{total characters} \times \text{average length})$. A single boolean `is_end` marks the end of a word while the shared path still holds each character exactly once. If you do need the actual string (e.g. to return it), reconstruct it from the path during the walk - that's what `prefix + acc` does in the DFS above - rather than storing copies.